

KrishiSetu.

PROBLEM STATEMENT ID	SIH26033
PROBLEM STATEMENT TITLE	Farmer-first agricultural trading platform with pooled delivery
THEME	Agriculture, FoodTech & Rural Development
PS CATEGORY	Software
TEAM ID	SIH26-SW059
TEAM NAME	Logic_Lords, Ramrao Adik Institute of Technology



“The farmer names the price. Everything after it is shared transport.”

Working prototype · krishisetu-weld.vercel.app

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KrishiSetu — pooled delivery

The farmer’s rate is an input to the price, never the leftover

TRANSPARENT COST COMPOSITION PER KG

Total 100%



●

Farmer ₹34.00

●

Transport ₹4.70

●

Packing ₹3.96

●

Site fee ₹3.00

₹45.66

BUYER PAYS

vs ₹55.00 quick-commerce

PRODUCER PAYOUT LEDGER • KHALAPUR & PANVEL CLUSTER

Sanjay Patil • Khalapur Raigad	3.0 kg /	₹102.00
Meena Bhoir • Panvel Rural Raigad	2.0 kg /	₹68.00

Net rate
nothing deducted, ever

Pooled by place
one neighbourhood, one vehicle

Sold before loaded
no stock written off

Fresher, cheaper
shorter chain, lower price

Demand by place
Nerul 246 • Kharghar 207

Surplus re-placed
151 kg split across three

FULFILLMENT AND LAST-MILE DISPATCH

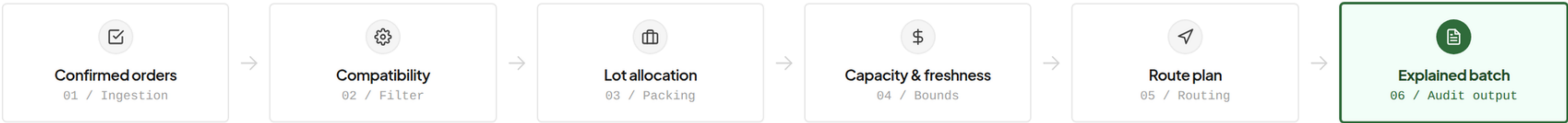
STANDARD Cluster pickup ₹10	DOORSTEP DROP Last 300 m ₹20	SINGLE-RIDER SPEED Express not offered
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Free above a ₹500 basket on the last 300 m. ₹55/kg is an assumed reference.

• Technical approach

D-COA — Decision & Coordination Optimisation Algorithm

Our own engine’s working name, not a claim of a new published algorithm.



EIGHT CONSIDERATIONS, EVERY RUN Feasibility first, then ranking

● Product compatibility	HARD	● Pickup & delivery geo	HARD
● Quantity vs capacity	HARD	● Delivery time windows	HARD
● Perishability & freshness	HARD	● Estimated transport cost	RANKING
● Order priority	RANKING	● Cancellations & new orders	REPLAN

● Hard constraint (zero-violation policy) ● Weight-ranked objective ● Dynamic event trigger



PRODUCTION STACK
Next.js 16 · PostgreSQL · Prisma · Vercel · live mandi prices, daily

RULES THE CODE ENFORCES

- 01 Under 70% fill, the run is held
No tempo leaves the cluster half-empty at producer expense.
- 02 Atomic reservations: no kilogram twice
Strict row-level locks prevent double-booking on pooled inventory.
- 03 A forecast shows only if it beats the baseline
Guards against erratic advisory outputs on farmer dashboards.

>

“Routing returns a feasible nearest-neighbour plan measured against a fixed-order baseline. We claim no global optimum.”

- **Feasibility and viability**

Return loads, the fill threshold, and what can go wrong



123

orders pooled

747.7 kg

on a 750 kg tempo

5

stops: 4 farms, 1 drop

₹5.37

transport per kg

One real run, planned by the live system

RISK, AND WHAT IS BUILT AGAINST IT

Too few orders

Held below 70% fill, rolls to the next window

Spoilage in transit

Freshness is a hard constraint, per commodity

Quality disputes

Graded, handover recorded, deduction by agreement

Farmer cannot use an app

Marathi, Hindi, English · one decision card

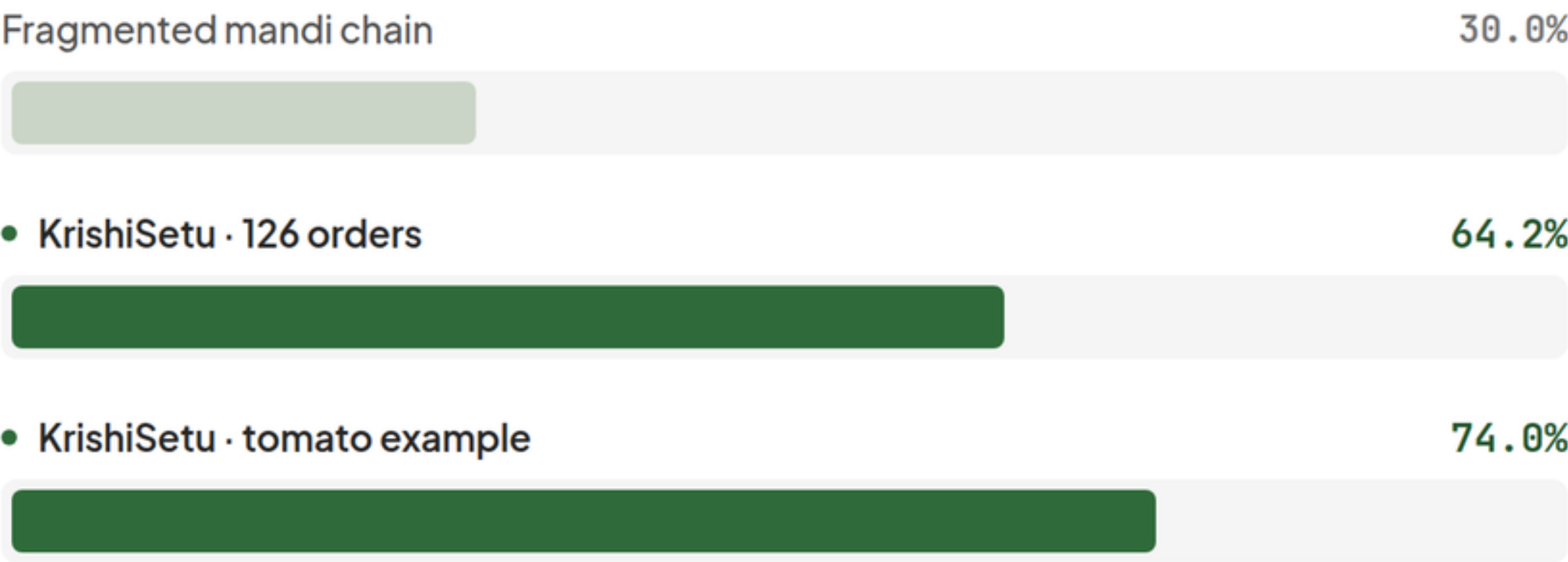
Return-load matching is designed and specified. It sits deliberately after the core proof and is not claimed as shipped.

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Impact and benefits

More for the farmer, less for the buyer, and still self-funding

FARMER'S SHARE OF WHAT THE BUYER PAYS



The fragmented-chain figure is a widely reported estimate. The two KrishiSetu bars are computed from orders in the running system.

Where our own money comes from	
Disclosed buyer-side fee, per order	₹19.89
Gateway, messaging, support	- ₹5.60
Contribution per order	₹14.29
Assumed monthly fixed cost	₹60,000
Orders a month to break even	4,199

No listing fee. No commission. To hold a price, our fee is cut to zero first — never the farmer's amount.

 Farmer ₹34.00 per kg, paid in full	 Buyer ₹45.66 per kg against a ₹55 reference	 Transporter ₹4,016 a run, agreed first	 Platform ₹14.29, one disclosed fee
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Figures are computed over synthetic demonstration orders: they size the model rather than measure a business.

• **Research and references**

Sources, and what they do and do not prove

 Government of India open data portal data.gov.in – daily mandi prices	LIVE IN APP	 eNAM — trading and FPO pages enam.gov.in/web/trading-details	INCUMBENT
 Ninjacart and Ninja Mandi ninjacart.com/ninja-mandi	INCUMBENT	 agribazaar agribazaar.com/quick-links/steps	INCUMBENT
 Samunnati samunnati.com/innovations	INCUMBENT	 Google OR-Tools, VRP with time windows developers.google.com/optimization/routing/vrptw	METHOD
 PIB release on agricultural marketing pib.gov.in – PRID 2222802	POLICY	 SIH 2026 idea submission format sih.gov.in/letters/2026	FORMAT

What this does not prove: demand history and the fitted price series are synthetic and labelled so on screen. An unmentioned capability on a competitor's page is not evidence they lack it, and we claim no measured superiority without a comparable pilot.